Program : **Diploma in Biomedical Engineering**	
Course Code : **6248**	Course Title: **Biomedical Instrumentation Lab II**
Semester : **6**	Credits: **4**
Course Category : **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester : **60**

Course Objectives:

- To provide hands-on experience in the circuit level knowledge of various therapeutic equipments.
- To enable the students to conduct case studies related to the hospital so that students will get idea for various duties of biomedical engineers.

Course Prerequisites:

Topic	Course code	Course name	Semester
Electronic Circuits		Electronic devices and Circuits	3
Op-amps and timer ICs		Linear Integrated circuits	4
Defibrillator, Pacemaker , ESU, Ventilator		Therapeutic Equipment	4
Ultrasound machine		Medical Imaging Techniques	5
Design of hospital rooms		Hospital Systems	5

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement internal circuits (AGC, ESU, pacemaker and X-ray timer) of biomedical equipment.	12	Applying
CO2	Demonstrate the operation of various therapeutic equipments viz. ESU, Defibrillator, Pacemaker and Ventilator	12	Understanding

CO3	Operate an ultrasound machine for getting a thorough knowledge about the equipment	5	Applying
CO4	Conduct case studies regarding the various facilities needed in a hospital to understand the role of biomedical engineer in the hospital.	13	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	2						
CO3	3	2					
CO4	3	3	3			2	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

_Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Design and Implement internal circuits (AGC, ESU, pacemaker and X-ray timer) of biomedical equipment.**		
M1.01	Implement an AGC circuit.	3	Applying
M1.02	Implement internal circuits X-ray timer of biomedical instruments	3	Applying
M1.03	Implement a pacemaker circuit and plot the output waveform.	3	Applying
M1.04	Implement an ESU circuit and generate waveform	3	Applying
CO2	**Demonstrate the operation of various therapeutic equipments viz. ESU, Defibrillator, Pacemaker and Ventilator**		
M2.01	Demonstrate the accessories, modes of operation and working of Electrosurgical unit.	3	Understanding
M2.02	Demonstrate the operation of an defibrillator/cardioverter	3	Understanding

M2.03	Demonstrate the operation of an pacemaker using a pacemaker simulator	3	Understanding
M2.04	Demonstrate the operation of a ventilator	3	Understanding
	Series Test - I	1.5	
CO3	**Operate an ultrasound machine for getting a thorough knowledge about the equipment**		
M 3.01	Identify the accessories of an ultrasound machine. Select a probe for a particular application	2	Applying
M 3.02	Demonstrate the various controls and modes of an ultrasound machine	2	Understanding
M 3.03	Conduct preventive maintenance of ultrasound machine	1	Applying
CO4	**Conduct case studies regarding the various facilities needed in a hospital to understand the role of biomedical engineer in the hospital.**		
M 4.01	Case study 1 : Design of X-ray /Audiometry/ Radiotherapy / CSSD/operation theatre room in a hospital with required size and other necessary requirements of the room after collecting required information from a nearby hospital.	5	Applying
M 4.02	Case Study 2: Conduct a study on procurement procedure of any one department viz. OT/Cardiology ICU/ Pediatric ICU/Surgical ICU/Emergency /Radiology departments - Prepare the list of equipment needed in the department - Collect specifications of the latest machines - Prepare an estimate - Prepare tender procedure - Prepare comparison statement - Prepare purchase order	5	Applying
M 4.03	Case Study 3: Design of electrical wiring in a ward/ICU/OT with emphasis of Earthing of hospitals after collecting details from a hospital. - Prepare schematic of wiring - Specify the procedure for hospital earthing.	3	Applying
	Lab Exam - II	1.5	

Text / Reference:

T/R	Book Title/Author
T1	Op-Amps and Linear Integrated Circuits”- Ramakant A Gayakwad - PHI- 4th Edition R2 “Handbook of Biomedical Instrumentation”- RS Khandpur- Tata McGraw-Hill Pub., 2006
R1	“Handbook of Biomedical Instrumentation”- RS Khandpur- Tata McGraw-Hill Pub., 2006
R2	G.D. Kunders - *Hospitals: Facilities Planning and Management,* Tenth edition, Tata McGraw Hill-India, 2008

Online Resources:

Sl.No	Website Link
1	www.allaboutcircuits.com
2	http://www.datasheetcatalog.com/
3	Ultrasound machine https://www.youtube.com/watch?v=7brAvZCg53E
4	Ultrasound knobology https://www.youtube.com/watch?v=qYJQWrSKjAY
5	Pacemaker https://www.youtube.com/watch?v=DHJT9eJXHLI
6	Webinar on OT design https://www.youtube.com/watch?v=xyNsU0uUkl4